

Tongyu Wang

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EDUCATION

University of Science and Technology of China (USTC)

BSc. Theoretical and Applied Mechanics | Minor in Computer Science

Hefei, China

Sept. 2023 - Jul. 2027

- **Overall GPA:** 3.68/4.3 (88.05/100)
- **Department Comprehensive Ranking:** 12/125 (Top 10%)
- **Awards:** USTC Talent Student Silver Scholarship | Top 7% | Twice for 2024 & 2025
- **TOEFL:** 106 | R28/L29/S23/W26

PERSONAL SITE

- [Tongyu\(Tory\) Wang](#)

PATENTS

- Yu Zhang, **Tongyu Wang**, Xiaohan Tang. Heavy-Load Gravity Energy Storage System. CN Invention Patent No. ZL 202511746905.8, Granted Jun. 26, 2026. Publication No. CN121251544B.
- Yu Zhang, **Tongyu Wang**, Xiaohan Tang. Transmission System for Gravity Energy Storage and Gravity Energy Storage System. CN Invention Patent Application No. 202522556214.3 (Patent Pending)
- Yu Zhang, **Tongyu Wang**, Xiaohan Tang. Mountain-Based Gravity Energy Storage System. CN Invention Patent Application No. 202511765995.5 (Patent Pending)

RESEARCH EXPERIENCE

Biomimetics & Dexterous Manipulation Laboratory(BDML) <https://bdml.stanford.edu/>

Stanford University

Core Researcher, supervised by Prof. Mark Cutkosky

Jun. 2026 - Present

Research Domain: Development of a Medical-Oriented Massage Dexterous Robotic System with Multimodal Sensing and VR Teleoperation

- Debugged an all-in-one circuit board integrating cameras, CoinFT sensors and LED strips, and implemented real-time pose tracking using Meta Quest 3S.
- Designed and fabricated the dome mold, created silicone domes, and tested optical performance.
- Designed a 2-DOF wrist mechanism, ran task-specific robotic-arm simulations in IsaacLab, and compared the motion performance of 7-DOF and 9-DOF manipulators under identical task conditions.
- Achieved real-time motion control of the Sharpa dexterous hand using the Meta Manus data glove.
- At present, I am working on deploying the TacClip sensor on the Sharpa hand for sensory-data acquisition, and implementing learned motion mapping to accomplish medical-relevant manipulation tasks.

Biomedical Mechatronics (BM2) lab, Georgia Institute of Technology

Remote

Research Assistant, supervised by Prof. Yue Chen and PhD student Jia Shen

Apr. 2026 - Present

Research Domain: Shape-Environment Fusion-Based Force Sensing for Continuum Robots

- Developed a multi-modal force-sensing framework for continuum robots that fuses shape and environmental sensing, resolving the numerical ill-conditioning inherent in purely shape-based methods.
- Formulated a high-fidelity contact modeling framework by integrating Coulomb friction complementarity constraints into the Cosserat rod model.
- Programmed a constrained Extended Kalman Filter (EKF)/MAP estimator to dynamically fuse simulated FBG curvature measurements with vision data, optimizing the posterior state subject to contact complementarity constraints.
- Engineered a closed-loop simulation pipeline integrating a forward Cosserat rod model (force-to-shape) with the proposed inverse estimator (shape-to-force). Benchmarked the algorithm against multiple state-of-the-art baselines, rigorously verifying its superior accuracy in reconstructing distributed forces.
- Established experimental validation protocols utilizing a physical catheter robot equipped with a stereo camera and FBG sensors, validating predicted contact forces against ground-truth data from sub-plane force sensors.

Soft Mechanics and Robotics Lab, USTC

Hefei, China

Researcher, supervised by Prof. Liu Wang, USTC

Oct. 2025 - Present

Research Domain: Research on Sim2Real Optimization of Magnetic Control System for Robotic Arms in Vascular Interventional Surgery

- Constructed a comprehensive ROS/Gazebo-based surgical simulation environment to model the kinematics and constrained workspace of a robotic arm, evaluating its capability to safely and precisely maneuver an external magnetic field generator around patients.
- Formulated a robust collision-avoidance and trajectory generation pipeline utilizing the OMPL library; leveraged the PRM* (Probabilistic Roadmap) algorithm coupled with Cartesian interpolation to compute smooth, time-efficient, and strictly collision-free paths.
- Developed a high-fidelity continuum robot simulation pipeline using the SOFA framework to model complex endovascular physics, accurately simulating real-time advancement, bending, and steering behaviors of a magnetic guidewire while accounting for continuous vessel-wall contact mechanics.
- Designed an autonomous Sim2Real control policy by integrating the SOFA physical simulation with a Reinforcement Learning (RL) framework, training an RL agent to optimize external magnetic field vectors for precise guidewire navigation based on real-time contact forces, localized friction, and tip velocities.

Physics of Fluids Lab, University of Twente

Enschede, Netherlands

Researcher, supervised by Prof. Corinna Maass, University of Twente

Jul. 2025 - Sept. 2025

Research Domain: Symmetry Breaking and Swimming Dynamics of Active Drops in a Surfactant Solution

- Mastered fluid dynamics expertise and grasped physical analysis methods for droplet self-propulsion within one week.
- Investigated the self-propulsion dynamics of micro-droplets, focusing on motion pattern transitions under high Péclet numbers to validate theoretical critical Péclet numbers and velocity thresholds.
- Executed CFD numerical simulations using C++ based Basilisk software, and developed and ran droplet motion simulation codes, generating motion demos and flow field visualizations.
- Processed simulation data via Python, analyzed and solved physical equations, and verified the correctness of theoretical models.

INTERNSHIP EXPERIENCE

Beijing Shidai Heavy Storage Technology Co., Ltd.

Beijing, China

R&D Intern

Sept. 2025 - Present

Responsibilities: Optimization of Mountain Slope-Type Ultra-High Power Gravity Energy Storage Cycle Transmission Device

- Derived unequal-height catenary equations to establish the core mathematical model for mountain-slope gravity energy storage systems.
- Formulated numerical solvers for transcendental equations to optimize chain slackness and calculate maximum sag under physical constraints.
- Developed MATLAB simulation protocols to validate the mathematical model, resolving non-analytical boundary conditions and pulley tangent point kinematics.

COURSE DESIGN PROJECT

Matchbox: Intelligent Wearable Networking System | Leader

Sept. 2025 - Jun. 2026

- Spearheaded the end-to-end development of Matchbox, an AI-powered wearable networking device featuring an e-ink display (MagTag) and wristband peripheral (M5Stick), utilizing ESP-NOW local wireless protocols for RSSI-based proximity sensing independent of venue Wi-Fi.
- Architected a robust MariaDB-backed RESTful API and React-based PWA for user onboarding, integrating an LLM (Gemini) to evaluate free-form interest profiles and autonomously generate tailored conversation prompts within a strict 5-second end-to-end delivery budget.
- Engineered a deterministic priority routing algorithm with multi-tier fallback (Peer-to-Peer, Exhibitor Lead, and Interest Suggestion), ensuring zero server-side failures during stress testing of 18 concurrent active users.
- Managed physical product delivery for the Stanford EXPE exhibition, including CNC mold-making, fabrication of 25 injection-molded hardware enclosures, and coordinating an Event-as-a-Service (EaaS) deployment for over 150 attendees.

EXTRACURRICULAR ACTIVITIES

- USTC Basketball Team | Member | Achievements: Top 4 in “Fire Phoenix Cup”, “Freshman Cup”; Top 8 in “Kexing Cup”
- USTC Cycling Association | Publicity Department Member | Managed video editing and social media outreach (Douyin, Bilibili)
- Piano: Level 10 Certification (China Musicians Association)

TECHNICAL PROFICIENCIES

- Coding Language: Python (2yr), C/C++ (2yr), MATLAB (2yr), LaTeX (Skilled)
- Tools: SolidWorks (2yr), Abaqus, COMSOL, Blender (3D Modeling), ROS, Mujoco, Isaac, Finite Element Analysis (FEA)
- Language Skills: English (Fluent), German (Elementary)